(Please write your Enrolment No. immediately)

Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Fourth Semester, May 2023

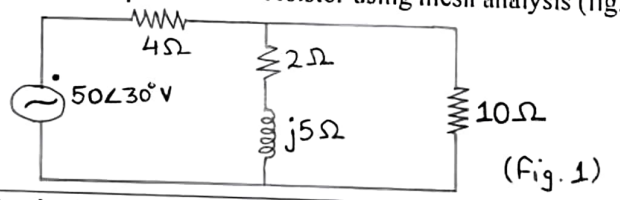
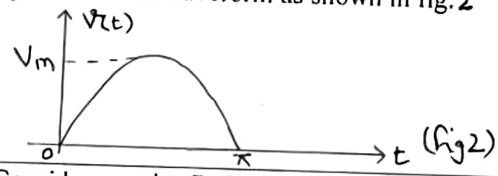
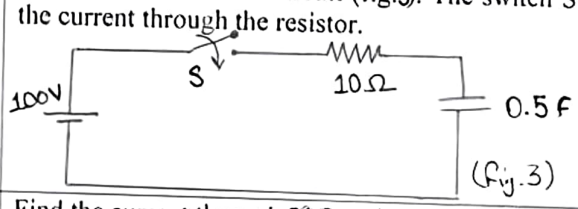
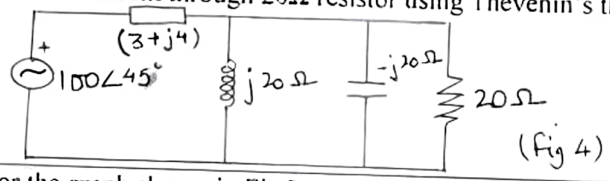
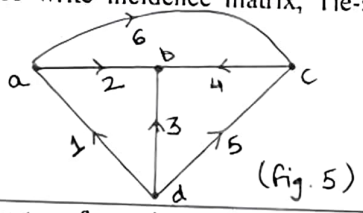
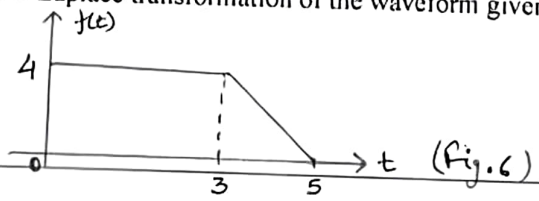
Paper Code: EEC-206

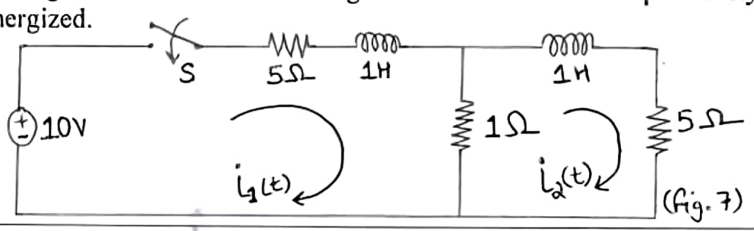
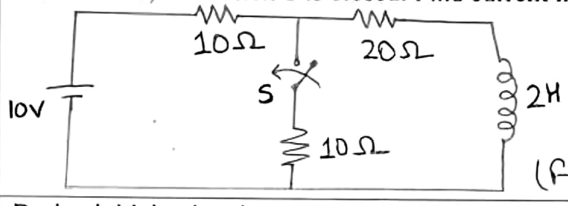
Subject: Network Analysis and Synthesis

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 a)	Explain tree and Co-tree with example	2	1
1 b)	Find the drop across 4Ω resistor using mesh analysis (fig. 1).  (Fig. 1)	3	1
1 c)	Synthesize the waveform as shown in fig. 2  (fig2)	2	2
1 d)	Consider a series R-C circuit (fig.3). The switch S is closed at t=0, Find the current through the resistor.  (fig.3)	3	2
2 a)	Find the current through 20Ω resistor using Thevenin's theorem (Fig.4)  (fig 4)	6	1
2 b)	For the graph shown in Fig5, consider the tree formed by branches (2, 3, 4), using this tree write incidence matrix, Tie-set matrix and cut-set matrix.  (fig. 5)	4	1
3 a)	Obtain the Laplace transformation of the waveform given in fig.6  (fig.6)	5	2

3 b)	Using Laplace transformation technique, find $i_2(t)$ at $t = 0^+$ following switching at t=0 of switch S in fig.7. Assume the network previously deenergized.  (fig. 7)	5	2
4 a)	In the circuit (fig.8), the switch S is open and the circuit reaches a steady state. At t=0, the switch S is closed. Find current in the inductor for t >0.  (fig.8)	6	2
4 b)	Derive initial value theorem and final value theorem.	4	1